
AUTOMATED FABRIC DEFECT DETECTION FOR TEXTILE MILLS

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ABSTRACT

The textile industry demands high-quality fabric output with minimal defects. Manual inspection is time-consuming, error-prone, and unscalable. This report presents an automated fabric defect detection system leveraging YOLOv11 — a state-of-the-art object detection architecture — trained on 20 distinct defect classes. The system captures fabric images, preprocesses them, runs real-time inference, stores structured results, and visualizes them through an interactive Power BI dashboard for quality monitoring and reporting.

1. INTRODUCTION

Fabric defects — holes, stains, weaving errors, surface irregularities — significantly affect commercial value in textile manufacturing. Traditional manual inspection is subjective, fatiguing, and error-prone at scale.

With advances in deep learning, automated visual inspection has become practical. CNN-based detectors, particularly the YOLO family, offer real-time detection with high accuracy. This project leverages **YOLOv11**, the latest generation, to build an end-to-end automated defect detection pipeline covering:

- High-resolution image acquisition
- Image preprocessing and augmentation
- YOLOv11 multi-class defect detection
- Confidence-scored classification
- Structured result storage (CSV/DB)
- Power BI visualization and alerting

2. DATASET DESCRIPTION

The dataset is a curated fabric defect dataset from Roboflow Universe, containing high-resolution images collected under controlled industrial lighting. It spans 20 defect classes covering structural weaving defects and surface contamination.

Parameter	Value
Total Images	846
Defect Classes	20
Annotation Types	Bounding Box & Smart Polygon
Image Format	JPEG / PNG
Source	Roboflow Universe
Project Type	Object Detection (Multi-label)
Min. BBox Size	10×10 px
Target IoU	≥70%

Images were collected using industrial-grade cameras on fabric inspection rigs. Multiple fabric types (plain weave, twill, satin) were included. The dataset is split 70% training, 20% validation, 10% test using stratified sampling.

3. THE 20 DEFECT CLASSES

Bounding Box Defects (13 Classes)

Used for compact, localized defects with roughly convex shapes.

#	Defect	Category
1	Hole	Structural
2	Slub	Yarn
3	Knots	Structural
4	Piling	Surface
5	Spot	Contamination
6	Fly Yarn	Contamination
7	Miss Pick	Weaving
8	Weft-Bar	Weaving
9	Warp-Bar	Weaving
10	Press-Off	Structural
11	Looseness	Weaving
12	Misprinting	Printing
13	Abrasion Marks	Surface

Key Bounding Box Defects:

Hole — Physical perforation where threads are missing; among the most critical defects as they render fabric unusable. Appear as dark circular or irregular voids.

Miss Pick — One or more weft threads absent from the weave, creating a visible horizontal gap. Weakens structural integrity; caused by loom timing issues.

Weft-Bar / Warp-Bar — Horizontal or vertical banding caused by irregular tension or density variation in weft/warp yarns respectively.

Spot — Localized stain or discoloration from oil, grease, dye bleeds, or chemical contamination. Highly variable in color, shape, and size.

Smart Polygon Defects (7 Classes)

Used for defects with irregular, elongated, or complex boundaries.

#	Defect	Category
14	Crease	Handling
15	Open Seam	Structural
16	Seam Puckering	Finishing
17	Damaged Selvedge	Structural
18	Snag	Surface
19	Skip Stitch	Stitching
20	Mis-threaded	Weaving

Key Polygon Defects:

Crease — Sharp fold lines pressed into fabric during processing or storage, creating visible distortion lines.

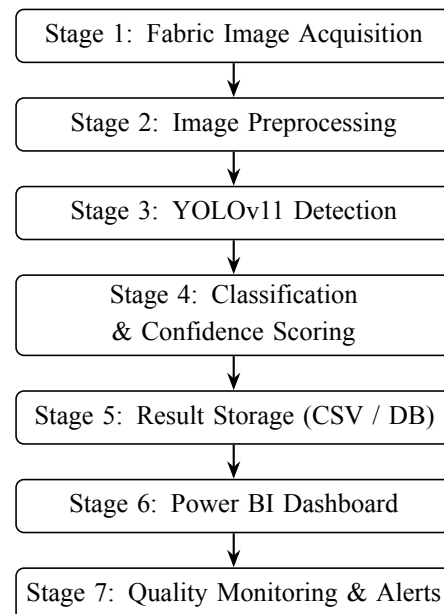
Seam Puckering — Irregular gathering along a seam caused by thread tension imbalance or differential shrinkage.

Skip Stitch — Missed stitches creating a broken pattern; weakens seams and reduces aesthetic quality.

Mis-threaded — Threads incorrectly routed through heddle/reed positions, causing repeating vertical pattern anomalies across fabric length.

4. SYSTEM ARCHITECTURE

The system follows a linear seven-stage modular pipeline. Each stage can be independently upgraded.



Hardware: Industrial cameras (≥ 5 MP), NVIDIA RTX GPU server, structured LED lighting, NAS for

archiving.

Software: Python 3.10+, Ultralytics YOLOv11, OpenCV, Pandas/SQLite, Power BI Desktop, Roboflow.

5. WORKFLOW EXPLANATION

Stage 1 — Image Acquisition: Industrial cameras continuously capture fabric at 30–60 fps. Structured LED lighting ensures uniform illumination. Images are transmitted via GigE Vision protocol.

Stage 2 — Preprocessing: Raw images are resized to 640×640 px; converted BGR→RGB; contrast-enhanced via CLAHE; noise-reduced with Gaussian blur; pixel values normalized to [0, 1]. Training augmentations include flipping, rotation, mosaic, and random cropping.

Stage 3 — YOLOv11 Detection: Pre-trained weights fine-tuned on the fabric dataset. Single-pass detection outputs bounding boxes, class labels, and confidence scores. NMS eliminates duplicate detections. Operates at >30 fps on GPU.

Stage 4 — Classification & Scoring: Each detection carries a class label (1 of 20) and confidence score (0–1). Confidence threshold: 0.5. Severity tier (*Minor* / *Moderate* / *Severe*) assigned by bounding box area relative to image area.

Stage 5 — Result Storage: Records stored in CSV and SQLite/PostgreSQL. Each entry includes: filename, timestamp, defect class, bounding box coordinates, confidence, severity, and pass/fail flag. Annotated images saved for human review.

Stage 6 — Power BI Dashboard: Live-connected or scheduled refresh from storage. Displays: defect count by class, defect rate trends, shift/machine comparison, fabric roll heatmap, pass/fail metrics, and exportable PDF/Excel reports.

Stage 7 — Quality Monitoring: Automated alerts (email/SMS) when defect rate exceeds threshold. Daily shift-based summary reports auto-generated and distributed. Insights drive loom parameter tuning and maintenance scheduling.

REFERENCES

- [1] Redmon, J. et al. (2016). You Only Look Once. *CVPR 2016*.
- [2] Ultralytics. (2024). YOLOv11. ultralytics.com
- [3] Roboflow Universe. Fabric Defect Dataset. universe.roboflow.com
- [4] Zhang, X. et al. (2023). Deep Learning for Textile Defect Detection. *IEEE Trans. Ind. Informatics*.
- [5] OpenCV Documentation. opencv.org
- [6] Microsoft Power BI Docs. powerbi.microsoft.com